6 Months Training TR-104 Full Stack Development

February, 2026

Day 13 – Authentication Service Optimization and Centralized Logic

1. Objective of the Day

The objective of the day was to optimize the authentication system by centralizing logic using services and improving code structure for better maintainability.

2. Tasks Performed

- Reviewed existing authentication implementation
- Refactored authentication logic into a centralized service
- Ensured all authentication operations are handled through a single service
- Improved code structure for better reusability
- Implemented methods for login, logout, and authentication state retrieval
- Removed redundant code from components
- Ensured consistent usage of authentication service across the application
- Tested functionality after refactoring

3. Tools / Technologies Used

- Angular Framework
- Firebase Authentication
- AngularFire
- TypeScript

4. Key Learnings

- Centralized services improve code maintainability

- Separation of concerns leads to cleaner architecture
- Services act as a single source of truth for application state
- Refactoring enhances scalability of applications
- Clean code improves readability and debugging

5. Challenges Faced

- Refactoring existing code without breaking functionality
- Ensuring consistent integration across components

6. Solutions

- Refactored code step-by-step
- Tested each feature after modification

7. Conclusion

- Successfully optimized authentication logic using services
- Improved overall application structure